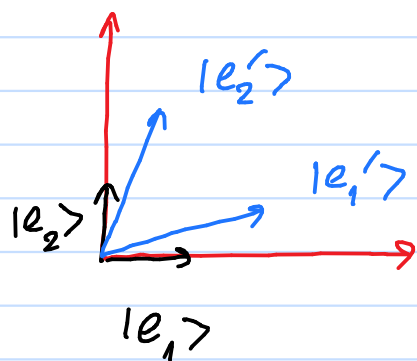


Symmetry Method Homework 3

1) Lorentz transformation

Consider a transformation on 2D vector space basis



$$\text{such that } \langle e'_1 | e_1 \rangle = \cosh \eta$$

$$\& \langle e'_1 | e_2 \rangle = \sinh \eta$$

$$\text{and } \langle e'_2 | e_2 \rangle = \cosh \eta \& \langle e'_2 | e_1 \rangle = \sinh \eta$$

1.1) Write down this basis transformation operator in the matrix form $\langle e_i | \hat{U} | e_j \rangle$

1.2) Consider the flip operator $\hat{T} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

How does this change under the basis transformation $\hat{U}$?

2). $\mathbb{Z}_2$ representation

Consider $\mathbb{Z}_2$ group = $\{1, K\}$

2.1) Find the regular representation of $\mathbb{Z}_2$

2.2) Since we know that an abelian group has only 1D irreducible representations, find the similarity transformation that brings the regular representation to the block diagonal form.